MILITARY SPECIFICATION SHEET

BATTERY, RECHARGEABLE, SEALED NICKEL CADMIUM, BB-593/U

This specification sheet is approved for use within Laboratory Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the sealed nickel cadmium rechargeable battery described herein shall consist of this document and the latest issue of MIL-B-49436.

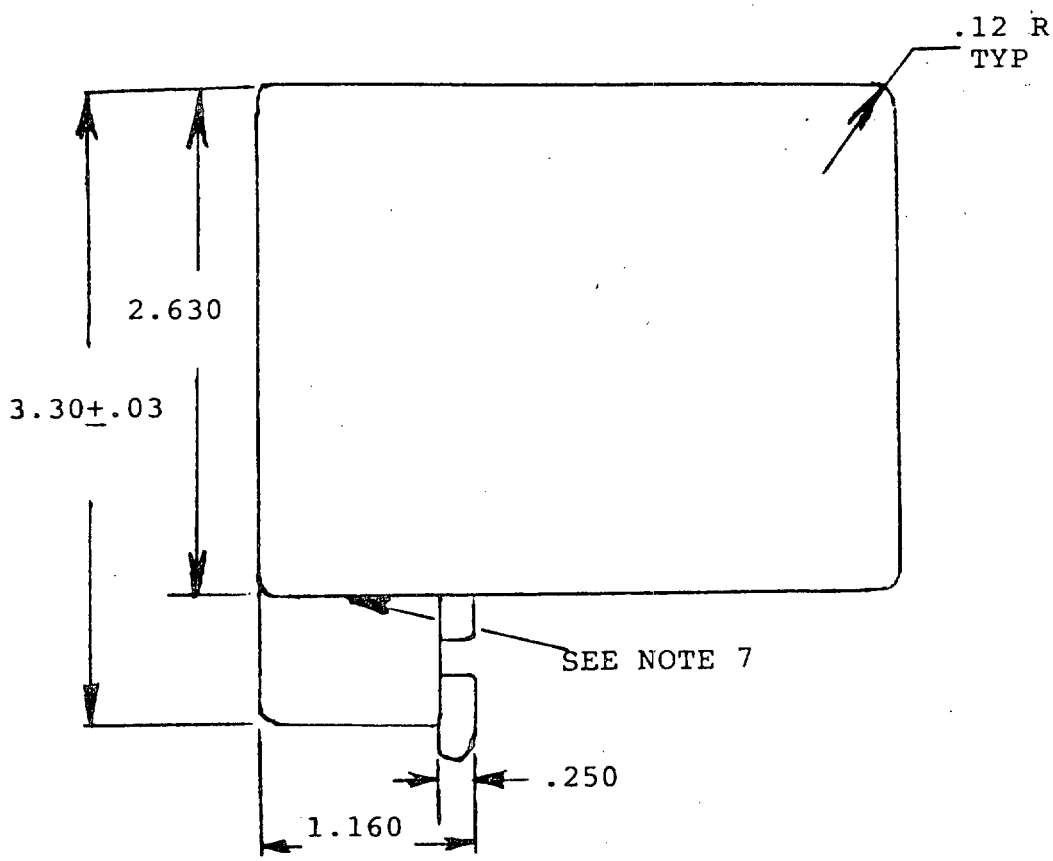


Figure 1. Side view of BB-593/U battery.

AMSC N/A

FSC 6140

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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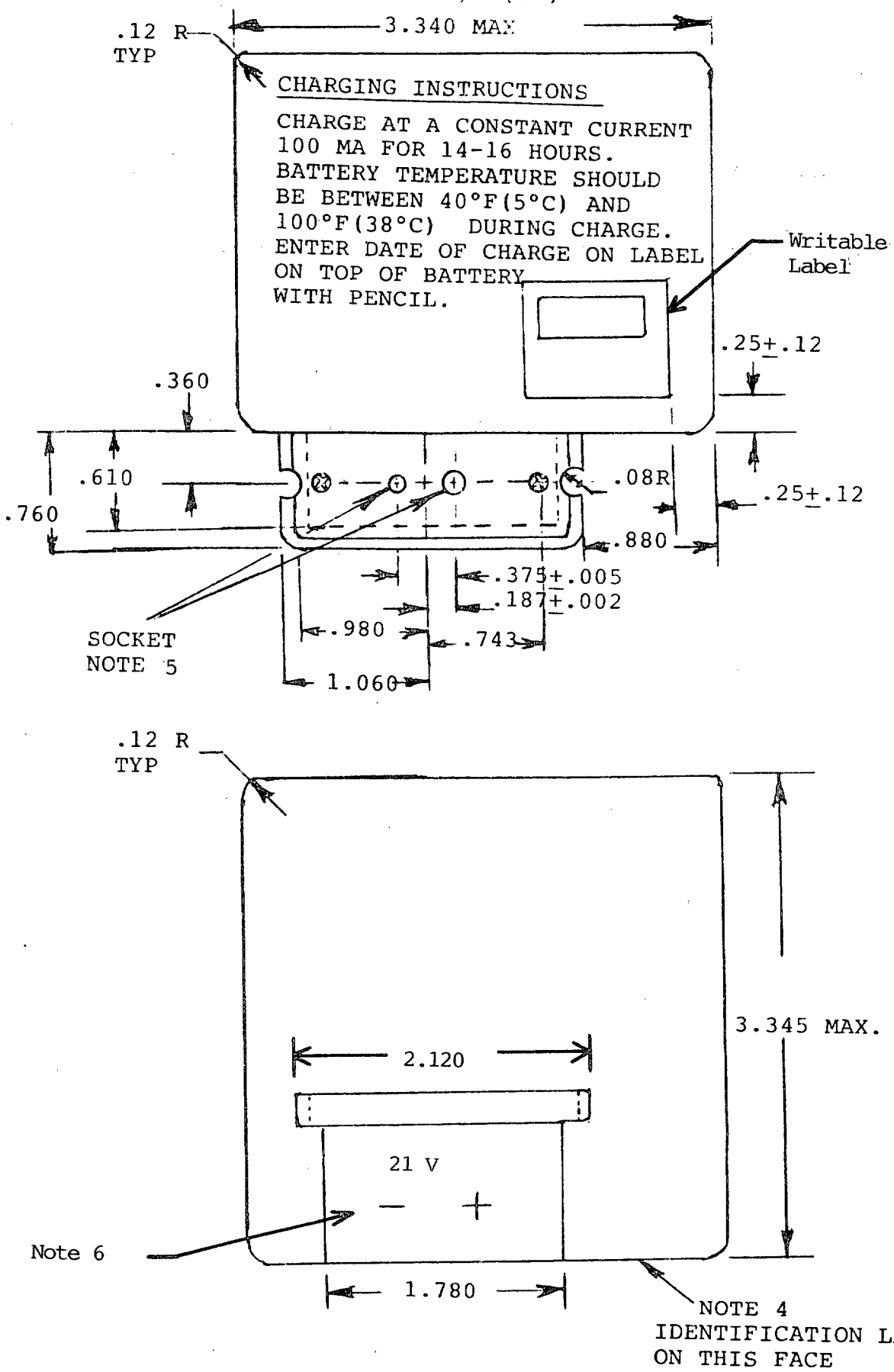


FIGURE 2: TOP AND FRONT VIEW OF BB-593/U BATTERY.

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DRAWING NOTES

1. All dimensions in inches.

2. Tolerances on decimals: ±0.015 inch, unless otherwise specified.

3. Dimensioning shall be per USAS-114.5 Y14.5 1966.

4. Stencil (or equivalent) identification and charging instructions of Figure 2. Manufacturers must fill in applicable instructions in parenthesis. Marking color shall be white.

5. The battery connector shall be a 2 pin socket connector type IV per MIL-B-18. The battery connector shall be capable of withstanding a minimum of 500 insertions and withdrawals with its mating connector without evidence of breakage, distortions, or loss of electrical performance. The socket shall be plated to prevent corrosion.

6. The "+" and "-" signs shall be raised 1/16 inch and be 1/4 inch in size.

7. A pressure relief mechanism shall be located in the case, with the preferred location under the connector. It shall release gas before case damage is incurred.

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REQUIREMENTS:

Rated Capacity (C_5)	1.0 AH
Voltage (nominal):	21.5 volts
Weight (maximum):	0.9 kg (2.0 lb)
Connector:	Two hole socket terminal per MIL-B-18, Type IV
Number and type of cells:	18 sealed "3/5 C_S" size cells or equal in series
Charge rate for testing and on battery (C/10 for 14 hrs.)	.10 amps

Discharge Rates	5 Second Voltage (V)	Final Voltage (V)	Current (Amps)
C/5	----	18.0	.200
$1C_5$	----	18.0	1.0

Test	Discharge Rate	Capacity Requirements (AH)
Full Capacity Discharge	$1C_5$	0.85
Cycle Life, 224 cycles	$1C_5$	0.68
Overcharge, 24 hours	C/5	1.0
High Rate Discharge	Not applicable	----
Low Temperature Discharge at -17°C (0°F)	C/5	0.8
Retention of Charge 7 days at 51°C (124°F)	$1C_5$	0.42
Pulse Discharge	Not applicable	----
Vibration	C/5	1.0

NOTES:

1. Unless otherwise specified, all values are minimum.
2. All charges and discharges shall be performed on fully assembled batteries through the terminals.

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The identification label shall be located on the bottom face of the battery. The following information shall be on the label:

BATTERY RECHARGEABLE BB-593/U
NICKEL-CADMIUM SEALED
21.0 VOLTS 1.00 AMP-HOURS AT 0.2 AMPS
NSN ( )
(CONTRACT NUMBER)
(DATE CODE) (SERIAL NUMBER)
(MANUFACTURER)
(LOCATION)

Information within the parenthesis shall be filled in by the manufacturer, as applicable.

The following requirements/tests of MIL-B-49436 are not applicable or are revised for the BB-593/U:

REQUIREMENT:

a) Mounting screws for connector: two (2) mounting screws shall be furnished with the battery connector. They shall be corrosion resistant steel, class 303 condition A per ASTM A484. Mounting screws shall be tightened so as not to impede any free movement of the socket connector.

b) Terminal markings: Schematic drawing is not applicable.

c) Electrical leakage: Not applicable.

TESTS:

a) Cycle life test: Except that the electrical leakage is not applicable.

b) Battery case vent test: Fixture shall not be required. The battery shall meet the requirement as described in paragraph 3.4.2 of the basic specification.

Custodian:
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Preparing Activity:
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(Project 6140-A658)

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